A Glacier's Warm Water Bath

*by **Tim Bartholomaus**, Geophysical Institute and Department of Geology and Geophysics, UAF*

The tidewater glacier cycle

In the mid-1970s, Alaska's ocean-terminating glaciers, or tidewater glaciers, helped scientists define the "tidewater glacier cycle," the dramatic pattern of advance and retreat exhibited by this type of glacier (Post et al., 2011). The advance portion of this cycle was inferred from geologic evidence to persist for centuries, with the glacier's toe, or terminus, creeping forward at rates of 1–10 m/yr. However, retreat was observed to be far more rapid. Glaciers could execute substantial, 10-km retreats in just a matter of decades.

Since its description, the tidewater glacier cycle has been of great interest to glaciologists and geophysicists seeking to understand glacier behavior, and has been the focus of careful observational and theoretical work. Worldwide, the largest and most rapidly changing glaciers and ice sheets end at tidewater. Understanding the mechanisms of the tidewater glacier cycle is essential if we hope to predict how these glaciers will continue to adjust in the face of global climate change. However, one of the most fundamental pieces of the puzzle, how and why tidewater glaciers lose mass from their termini, remains poorly understood. Work supported by the Center for Global Change (CGC) is helping us illuminate some of the processes that have long been obscured beneath the turbid, ice-choked, sea surface of Alaska's glaciated fjords.

Mass loss from a tidewater glacier terminus is known as frontal ablation. When frontal ablation rates increase at a glacier, a near-irreversible set of glacier wasting feedbacks can begin. The associated acceleration in ice flow and thinning is relatively well understood by glaciologists. However, the study of frontal ablation remains at glaciology's cutting edge.

From the earliest human visits to tidewater glaciers, people have described iceberg calving. Often spectacular, the sudden collapse of hundreds of thousands of tons of glacier ice is an obvious example of frontal ablation. Calving can be so astounding that, until just a decade ago, calving was frequently considered the only frontal ablation mechanism. This viewpoint changed when a study at a southeast Alaska glacier led by researchers from the UAF Geophysical Institute demonstrated that melting below the water line of a glacier (submarine melt) can be just as important as iceberg calving in dictating the frontal ablation rate (Motyka et al., 2003). However, recognition of the importance of submarine melt has been slow in coming and no subsequent estimates of submarine melt rates at Alaskan glaciers have been carried out.

Looking beneath the fjord surface

Icy Bay is a 40-km-long indentation in a remote part of the Gulf of Alaska and an excellent system in which to study the frontal ablation of tidewater glaciers *(Figure 1)*. Four significant glaciers, including Yahtse Glacier *(Figure 2)*, calve icebergs and disgorge their turbid runoff into fjords

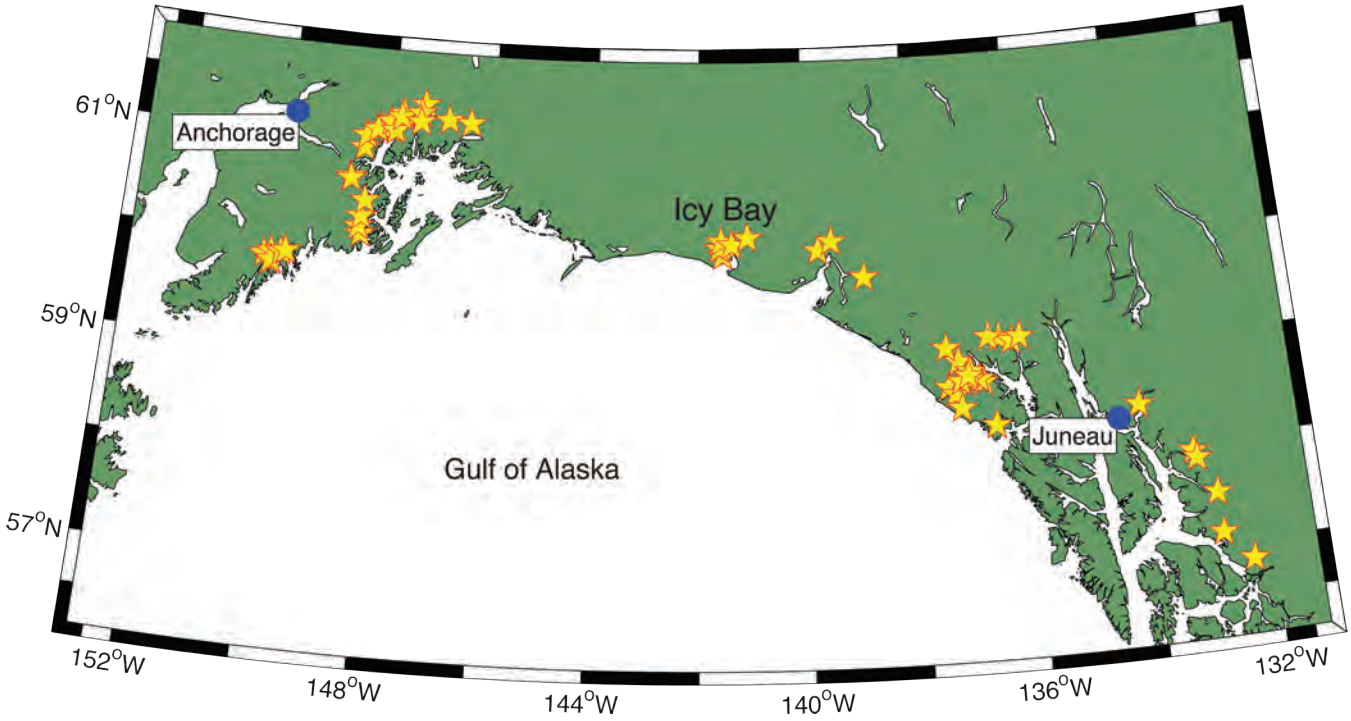


Figure 1. Gulf of Alaska region, with location of Icy Bay. Stars mark the locations of all tidewater glaciers in Alaska, most of which have undergone significant retreat over the last several centuries (adapted from Molnia, 2008).



Figure 2. Yahtse Glacier flows from the high mountains of the St. Elias Range to tidewater in Icy Bay. Frequent, spectacular iceberg calving from above the water line covers the sea surface in the foreground with floating "bergy bits." However, nearly all of the ice below the sea surface is lost to melt by warm ocean water. Photo by Tim Bartholomaus.

that branch from the head of Icy Bay. To tease apart the quantity of submarine melt in Icy Bay from the bay's other constituents (Gulf of Alaska seawater and fresh, subglacial discharge), I made two CGC-supported oceanographic trips to this part of the Alaskan coast, between the towns of Yakutat and Cordova. During these trips, I measured 117 vertical profiles of water temperature and salinity to characterize the water properties and circulation within the bay *(Figures 3 and 4)*. In every one of our profiles, we found a surface layer of cold (1–4 °C), relatively fresh (salinity of < 25 g/kg) water, 10–20 m thick. Below that, our profiles revealed that warm (> 6 °C) water from the Gulf of Alaska was reaching all the way to the glaciated head of Icy Bay; in 2010, water at 40 m depth reached 10.4 °C (51 °F) at a location only 1.5 km from the Yahtse Glacier terminus. A small chip of ice floating in 51 °F tap water will surely not last long before it melts. Thus, the question is: might submarine



Figure 3. Tim Bartholomaus pilots a skiff between measurements of water properties in Icy Bay. The water surrounding the boat is turned brown by high concentrations of glacial silt discharged into the sea water. Photo by Megan O'Sadnick.

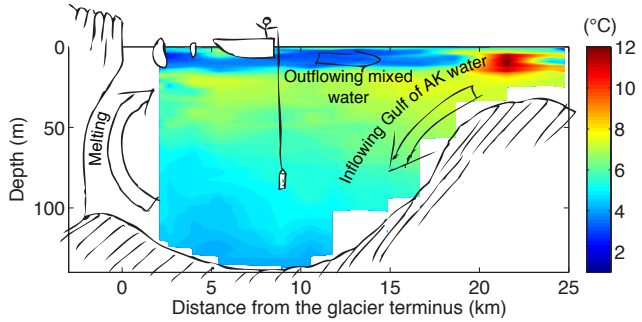


Figure 4. Tim measures the water temperature in a longitudinal cross section of Icy Bay, in front of the terminus of Yahtse Glacier. Water temperatures are those measured during July 2011. Warm, incoming Gulf of Alaska water enters Icy Bay over a submarine shoal, undercuts the glacier terminus by melting ice, then mixed melt, subglacial discharge and seawater flow out of the bay just beneath the surface.

melt at Yahtse Glacier be a significant portion of the frontal ablation rate? And, as we seek to understand the behavior of glaciers like Yahtse, do we need to consider the rate of ice melt as a process potentially as important as iceberg calving in producing ice loss?

Our desire to answer these questions pushed us to look more closely at our temperature and salinity data. If no submarine ice melt were to occur within Icy Bay, then only two sources of water would be mixing in Icy Bay—cold, fresh subglacial discharge and the warm, salty seawater from the ocean. In this simple model, any parcel of water would have a temperature and salinity that varied linearly depending on the relative proportions of the seawater–subglacial discharge mixture. However, our data told a different story, revealing that almost all of the water in the shallow, fresh layer is colder than would be expected from this simple model. The simplest way to account for this anomalous deficit in the water's sensible heat is the uptake of latent heat as glacier ice melts. Our calculations indicate that melted ice makes up 1–2.5% of the shallowest water. Field observations and timelapse movies of the iceberg circulation in front of Yahtse Glacier reveal moderate currents (8 cm/s) in these shallowest water layers. By combining the volume and current results, we derived a conservative melt rate of 9 m/day during late July at Yahtse Glacier, applied evenly over the entire submarine terminus. We know from other observations that the total frontal ablation rate at Yahtse Glacier is 19 m/day (equal to the rate of ice flow into the terminus). More mid-range estimates of the shallow melt water content indicate that submarine melt may easily account for all of the below-waterline frontal ablation.

Reinterpreting the drivers of tidewater glacier retreat

This view of frontal ablation is a major departure from the existing understanding. Many researchers seek physical laws that predict rates of mechanical iceberg calving based on ice stretching rates or fjord geometry. Our team's findings suggest that such efforts would be misplaced at Yahtse Glacier. Instead, calving may simply be a response to undercutting

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of the highly fractured glacier tongue. The glacier melts so rapidly below the water line that the broken ice blocks above simply topple in once they lose their pedestals.

Our oceanographic work supports an evolving perspective on the tidewater glacier cycle and indicates that high rates of submarine melt may be one of the key drivers of rapid tidewater glacier retreats. If a glacier terminus recedes into a deepening embayment, melt rates may increase when warm water gains access to more submarine ice. This might help trigger the catastrophic tidewater glacier retreats identified throughout the Gulf of Alaska during the last couple of centuries. The increasing awareness of the magnitude of submarine melt rates will likely help guide the design of new global change–focused research on the tidewater glacier cycle.

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Tim Bartholomaus, a Ph.D. student in Geophysics, received his CGC award in 2010.

Walrus Predation on Seals: Occurrence and Relationships with Arctic Sea Ice and Disease

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Background

Pacific walruses (*Odobenus rosmarus divergens*) are of cultural and dietary significance to Arctic Native subsistence communities, yet little is known about the ecology and current status of this species within the context of changing climate. The loss of sea ice, which provides a vital resting platform for walruses between foraging trips and a diving platform allowing access to clams, mussels, and other benthic (bottom-dwelling) prey, may prompt the walruses to adopt alternative foraging strategies (Rausch et al., 2007; Fischbach and Jay, 2008). These may take the form of direct changes, such as altered feeding behavior, or indirect changes in the form of differences in food web dynamics, decreased biomass input to the benthos, and decreases in prey quality. Arctic sea ice minima have been reported between 2007 and 2012, reducing this crucial habitat, and adversely impacting walruses' foraging success, as the ice edge retreats over waters too deep for the diving capacity of these pinnipeds.

While walruses are considered specialized benthic feeders with specially adapted facial musculature for consuming energy-rich bottom-dwelling mollusks (shellfish) (Fay, 1982), many other prey species (e.g., marine worms, snails, crustaceans) have been described from stomach content analyses (Sheffield et al., 2001), including seals and seabirds. Previous researchers (Fay, 1960; Rausch et al., 2007; Jay and Fischbach, 2008) proposed that under reduced ice extent or other nutritionally stressful situations, walruses may implement alternative feeding strategies, including prey-switching, focusing on seals and seabirds. Prey-switching is a common phenomenon of organisms in response to external factors, including declines in prey abundance. Increased reliance on alternate prey species can mitigate the loss of caloric energy from declines in availability of traditional prey. The use of centralized terrestrial haulouts in the absence of sea



Courtesy of Jill-Marie Seymour.

ice could also lead to localized prey depletion and/or create energetically costly increases in travel distances to foraging grounds (Sheffield and Grebmeier, 2009). Consumption of higher trophic level organisms by walruses is not a novel behavior, although the incidence of this foraging strategy appears to be increasing (Fay, 1960; Lowry and Fay, 1984; Wolkers et al., 2006).

The ability to quantify the importance of higher trophic level prey to walrus diet has historically been limited, as only the soft tissues (i.e., muscle, blubber) are ingested. These soft tissues are digested more readily than the tougher tissues of benthic invertebrates (e.g., clam siphons) (Sheffield and Grebmeier, 2009), decreasing the likelihood of positive identification of seal and seabird remains during examination of stomach contents.

It is well known that dramatic prey shifts (e.g., seal-eating) can alter host–parasite dynamics as well as result in declining body condition and fecundity, increased disease susceptibility, decreased offspring survival, and changes in contaminant exposure; all of which can lead to population declines (Burek et al., 2008). Walrus meat is already a known source of *Trichinella* spp., a parasite responsible for Trichinellosis outbreaks among Arctic Native subsistence communities. While it has been hypothesized that walruses contract Trichinella from eating infected seals, lack of accurate prevalence surveys in Alaska prevents confirmation of the parasite's transmission in the marine environment and